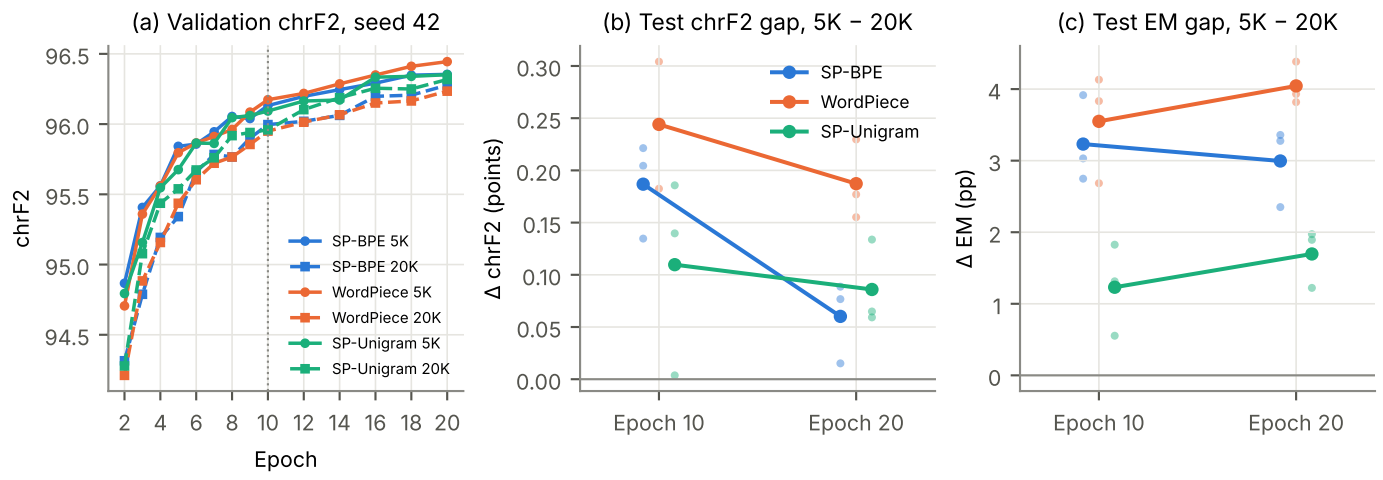
(a) Validation chrF2, seed 42
chrF2
96.5
96.0
95.5
95.0
94.5
2
4
6
8
10
12
14
16
18
20
Epoch
SP-BPE 5K
SP-BPE 20K
WordPiece 5K
WordPiece 20K
SP-Unigram 5K
SP-Unigram 20K
(b) Test chrF2 gap, 5K − 20K
Δ chrF2 (points)
0.30
0.25
0.20
0.15
0.10
0.05
0.00
SP-BPE
WordPiece
SP-Unigram
Epoch 10
Epoch 20
(c) Test EM gap, 5K − 20K
Δ EM (pp)
4
3
2
1
0
Epoch 10
Epoch 20